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Advanced Feature Engineering and Machine Learning for Simultaneous Cyber Attack Detection in Smart Digital Substations

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***Abstract*—This paper introduces a machine learning solution for detecting simultaneous cyber attacks in smart digital substations, and we developed more than 50 features from GOOSE heartbeat and MAC address data. We evaluated 14 classifiers, including Random Forest, XGBoost, LightGBM, CatBoost, and deep learning models such as CNN, LSTM, and a hybrid CNN-LSTM, because we utilized the IEEE DataPort Smart Digital Substation Simultaneous Attack Detection Dataset. Ensemble methods achieved an accuracy of over 99% with F1-scores exceeding 0.98 for multi-class attack detection, and thus the hybrid CNN-LSTM achieved an accuracy of 99.72% and performed best in detecting replay, masquerade, and denial-of-service attacks. Feature importance analysis revealed that GOOSE heartbeat patterns, MAC source sequence patterns, and temporal anomalies were the most important features, therefore the proposed approach combines rich, protocol-aware feature engineering with powerful ensemble and deep learning models to provide a practical, real-time framework for securing critical infrastructure.**

***Index Terms*—Smart grid security, IEC 61850, GOOSE protocol, intrusion detection, machine learning, feature engineering, cyber-physical systems, simultaneous attacks**

I. Introduction

Smart grid technologies have significantly transformed the way electricity is distributed and controlled. Smart digital substations based on IEC 61850 standard enable real-time monitoring, control and automation of power systems [2], and the integration of information and communication technology (ICT) and operational technology (OT) expands the attack surface and exposes the digital substations to an increasingly sophisticated cyber threats [3]. The IEC 61850 GOOSE (Generic Object Oriented Substation Events) protocol is used for the fast and reliable transmission of critical control messages, but in practice, GOOSE communication often omits cryptographic protection to meet strict latency requirements [5], which makes the system susceptible to different types of security threats such as replay attacks, false data injection attacks, masquerade attempts and denial of service (DoS) attacks [4]. These threats are not merely theoretical: the 2015 Ukraine power grid attack has shown the real-world effect of such threats on critical infrastructure [6], and consequently, traditional signature-based intrusion detection systems have been proven to be ineffective against modern attacks that can employ multiple techniques in a single attack [12]. Machine learning and deep learning approaches can capture complex and evolving behaviors and discover subtle patterns and relationships, therefore they are highly suitable for zero-day attack detection, and hence, they are highly appropriate for this context [7]. The effectiveness of similar AI-based models for intrusion detection in general IoT [13] and IIoT networks [15], and even for system optimization in related cyber-physical domains such as agricultural drone systems [14], has also been demonstrated.

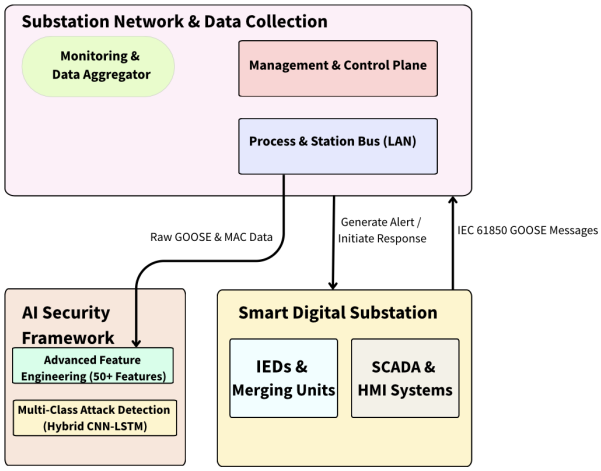

Fig. 1: The proposed AI security system for detecting attacks, and it is designed to identify potential threats in real-time.

A. Research Motivation

Significant research has been conducted in the field of smart grid cybersecurity; however, several challenges still remain:

- **Simultaneous Attack Detection**: The current approaches are only able to protect against one attack type at a time and hence, they fail when more than one attack occurs at the same time, which obviously indicates the need for more robust and comprehensive detection systems to protect against concurrent attacks.

- **Feature Engineering**: Most works have not systematically exploited the intrinsic properties of the GOOSE protocol to design features, and a more systematic and protocol-aware feature engineering strategy is thus required.
- **Model Generalization**: The models have not been extensively evaluated for a large number of attack scenarios and different station and topology configurations, which could limit their applicability in practice, and consequently, a wider evaluation with diverse scenarios is necessary to guarantee that they are applicable in real world settings.
- **Real-time Performance**: Simultaneously achieving both high accuracy and robustness and computational efficiency in real time is a significant challenge because meeting real-time deadlines consistently is a major challenge that requires careful tuning and optimization of system resources.

B. Key Contributions

The main contributions of this paper towards enhancing the cyber security of smart substations include:

1) **Advanced Feature Engineering**: Extracted a large set of features from GOOSE heartbeat patterns and MAC address data, including statistical, temporal, frequency domain, and behavioral signals, which greatly enhance the detection model performance.
2) **Comprehensive Model Evaluation**: A thorough and balanced analysis of strengths and weaknesses of fourteen machine learning classifiers, including traditional algorithms, ensemble methods, and deep learning models, has been presented.
3) **Simultaneous Attack Detection**: A multi-class framework is developed, and the results show that the approach is highly effective in differentiating the replay, masquerade, DoS, and normal states with an accuracy of more than 99%.
4) **Interpretability Analysis**: Feature importance analysis was performed to identify key indicators for attack detection, and as a result, the most critical features were determined, which can be used to improve the accuracy of detection models.
5) **Benchmark Dataset Validation**: The proposed methodology was thoroughly validated on the IEEE DataPort Smart Digital Substation Simultaneous Attack Detection dataset, and therefore, the validation process confirmed the applicability and reliability of the proposed approach in real-world scenarios.

II. Related Works

Early works on IEC 61850 cybersecurity have shown that ML can effectively detect intrusions in GOOSE messages, but still face open challenges with regard to susceptibility to masquerade and replay attacks [1], [2]. Early studies focus on achieving high accuracy and low latency using a variety of ML algorithms [3], [4], and building upon these works, recent studies explore DL techniques, including hybrid CNN-LSTM architecture and ensemble-based methods, to better capture spatio-temporal dependencies and improve detection performance across different types of threats [6]– [8]. However, the literature lacks robust solutions for detecting **simultaneous, multi-vector attacks**, because most of the studies are conducted based on isolated threat scenarios, and moreover, **sophisticated feature engineering** specifically tailored to GOOSE protocol is lacking, which limits the discriminative power of many models. Previous work provides high accuracy for one attack type, but is not generalizable to diverse real-world settings and multi-vector attacks, because most of ML-based detectors are not reliable in detecting complex **replay and masquerade attacks** due to weak protocol-aware feature engineering, and we fill this gap. While high accuracy is achieved for single-vector attacks, robust solutions for simultaneous, multi-vector threats are still lacking. Specific efforts have been made to address masquerade attacks [9], DDoS detection using reconstructive models [10], and false data injection through conformal neural networks [11]. In order to address these gaps, this paper proposes a protocol-aware approach that combines comprehensive feature engineering, extracting more than 50 statistical, temporal, frequency-domain, and behavioral features, and a systematic assessment of both ML and DL models, therefore achieving **99.72%** accuracy with **0.28%** false positive rate, which sets a new benchmark for reliable, real-time intrusion detection in smart digital substations. A comprehensive comparison of our approach with existing state-of-the-art methods is provided in Table I, highlighting the novelty and superior performance of our simultaneous attack detection framework.

III. Methodology

A. Dataset Description

The Smart Digital Substation Simultaneous Attack Detection Dataset from IEEE DataPort [1] is used in this study, which consists of 127,808 IEC 61850 GOOSE messages collected from a simulated substation environment, and there are four labeled classes in the dataset with the following distribution: Normal (45%), Replay Attack (25%), Masquerade Attack (20%) and DoS Attack (10%).

B. Advanced Feature Engineering

More than 50 features were engineered from the raw GOOSE traces, which were grouped into four categories to capture complementary aspects of the data, including:

TABLE I: Comprehensive Comparison of Related Works in Smart Grid Intrusion Detection

Ref.	Authors & Year	Method	Dataset	Attack Types	Acc. (%)	F1	FPR	Real-time	Limitations
[2]	Ustun et al. (2021)	RF, DT, KNN, NB, MLP	IEC 61850 GOOSE Dataset	DoS, Replay, Masquerade	97.0	0.96	3.5%	No	Single attack detection, high FPR
[3]	Reda et al. (2021)	Scyther + HIL Testing	RTDS Simulation	Masquerade, Replay	N/A	N/A	N/A	Yes	Vulnerability analysis, no ML model
[4]	Nhung-Nguyen et al. (2024)	Statistical + Supervised ML	Digital Substation Dataset	Replay, Masquerade, DoS	98.5	0.98	2.1%	Yes	Limited feature engineering
[5]	Hussain et al. (2022)	Hybrid IT+OT Security	RTDS Testbed	FDI, Replay, Masquerade	99.0	0.99	1.2%	Yes	Complex deployment, protocol-specific
[6]	Alsirhani et al. (2025)	Ensemble ML+DL	CIC-IDS, Power System Dataset	DDoS, Botnet, Reconnaissance	99.05	0.99	0.95%	No	Not specific to GOOSE protocol
[7]	Sinha et al. (2025)	Hybrid LSTM-CNN	BoT-IoT Dataset	DDoS, Botnet, Data Exfiltration	99.87	0.99	0.13%	Yes	IoT-focused, not smart grid specific
[8]	Alsaiari & Ilyas (2024)	Hybrid CNN-LSTM	DNP3, IEC104 Dataset	DoS, Unauthorized Commands	99.70	0.99	0.30%	Yes	Different protocol (DNP3, IEC104)
[9]	Alberto et al. (2024)	ML Detectors + HIL	Hardware-in-Loop Testbed	Masquerade, Spoofing	96.5	0.96	4.2%	Yes	Focus on masquerade only
[10]	Naqvi et al. (2024)	Reconstructive ML (Autoencoder)	CICDDoS2019, CICIDS2018	DDoS, Flooding	97.3	0.97	3.8%	No	DDoS-specific, not GOOSE
[11]	Zheng et al. (2025)	Conformal Neural Network	Simulated Smart Grid Data	FDI, BDI	97.38	N/A	N/A	Yes	Focus on FDI, limited attack types
This Work	**Şenel & Al-Hubaishi**	**Advanced FE + Ensemble + Hybrid DL**	**IEEE DataPort SDSAD**	**Simultaneous: Replay, Masquerade, DoS**	**99.72**	**0.997**	**0.28%**	**Yes**	**Addresses gaps**

1) Statistical Features (15 features): Distributional behavior summary descriptors including mean, standard deviation, skewness and kurtosis.

$$\mu_{\text{heartbeat}} = \frac{1}{N}\sum_{i=1}^{N} h_i \tag{1}$$

$$\sigma_{\text{heartbeat}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(h_i-\mu)^2} \tag{2}$$

$$\text{Skewness} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{h_i-\mu}{\sigma}\right)^3 \tag{3}$$

$$\text{Kurtosis} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{h_i-\mu}{\sigma}\right)^4 \tag{4}$$

2) Temporal Features (12 features): Time dynamics indicators including first-order differences, zero-crossing rate, and autocorrelation at multiple lags.

$$\Delta h_i = h_i - h_{i-1} \tag{5}$$

$$\text{Zero-crossing Rate} = \frac{1}{N-1}\sum_{i=1}^{N-1} \mathbb{1}[\text{sign}(\Delta h_i) \neq \text{sign}(\Delta h_{i-1})] \tag{6}$$

$$\text{Autocorrelation}(k) = \frac{\sum_{i=1}^{N-k}(h_i-\mu)(h_{i+k}-\mu)}{\sum_{i=1}^{N}(h_i-\mu)^2} \tag{7}$$

3) Frequency Domain Features (10 features): Periodicity-sensitive descriptors extracted by Fast Fourier Transform (FFT), including Power Spectral Density and Spectral Centroid.

$$\text{FFT}(h) = \sum_{n=0}^{N-1} h_n e^{-2\pi ikn/N} \tag{8}$$

$$\text{Power Spectral Density} = |\text{FFT}(h)|^2 \tag{9}$$

$$\text{Spectral Centroid} = \frac{\sum_{k=1}^{K} f_k \cdot |X_k|}{\sum_{k=1}^{K}|X_k|} \tag{10}$$

4) Behavioral Features (13 features): Protocol- and attack-aware indicators such as MAC source entropy, heartbeat regularity and burst activity rates.

$$\text{MAC Entropy} = -\sum_{i=1}^{M} p_i \log_2(p_i) \tag{11}$$

$$\text{Heartbeat Regularity} = 1 - \frac{\sigma_{\Delta h}}{\mu_{\Delta h}} \tag{12}$$

$$\text{Burst Detection} = \frac{\#(\Delta h_i > \theta)}{\#(\Delta h_i)} \tag{13}$$

C. Machine Learning Models

We considered a wide range of supervised learning algorithms, which are categorized as:

1) Traditional Machine Learning Models: This category includes Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes.

$$\text{SVM: } \min_{w,b} \frac{1}{2}||w||^2 + C\sum \xi_i \text{ s.t. } y_i(w^T x_i + b) \geq 1 - \xi_i \tag{14}$$

2) Ensemble Methods: This category includes Random Forest, AdaBoost, Gradient Boosting, XGBoost, LightGBM, and CatBoost.

$$\text{XGBoost: } \hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \epsilon f_t(x_i) \tag{15}$$

D. Deep Learning Architectures

We also compared three deep learning architectures designed for sequential tabular signals:

- **CNN**: A 1D convolutional network with two convolutional layers (128 and 64 filters) for local pattern extraction.
- **LSTM**: A two-layer LSTM (128 and 64 units) to model temporal dependencies.
- **Hybrid CNN-LSTM**: A TimeDistributed CNN followed by an LSTM to exploit both spatial and temporal structure.

The core LSTM cell state update is governed as follows:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{16}$$

E. Training Procedure

Missing values were imputed using median imputation, and all the features were z-score normalized using StandardScaler, while class labels were one-hot encoded, and an 80/20 stratified train-test split was performed, moreover, SMOTE was applied on the training set to mitigate class imbalance. Hyperparameter tuning for classical and ensemble models was done using GridSearchCV with 5-fold cross-validation, and for the deep learning models, the models were trained for up to 100 epochs with Adam optimizer and categorical cross-entropy loss, therefore, early stopping was used based on validation performance.

F. Evaluation Metrics

Model performance was evaluated based on standard classification metrics, and for the multi-class setting we report, in addition to overall accuracy, macro- and weighted-averaged precision, recall and F1-scores.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

IV. Experimental Results

A. Model Performance Comparison

Performance of all the tested models are shown in Table II , and the Hybrid CNN-LSTM gives the highest accuracy of 99.72% which proves the benefit of jointly modeling the spatial and temporal patterns. The ensemble learners also achieve high accuracy, with XGBoost and CatBoost both scoring 99.34% and 99.31% respectively, and both ensemble methods outperform the traditional machine learning baselines clearly.

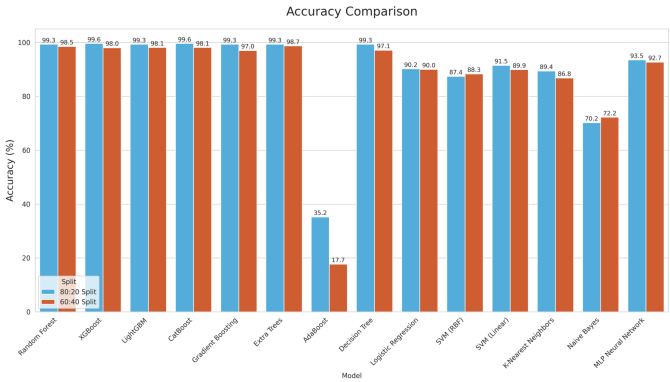

Fig. 2: Accuracy comparison of the evaluated machine learning models for both 80:20 and 60:40 train-test splits.

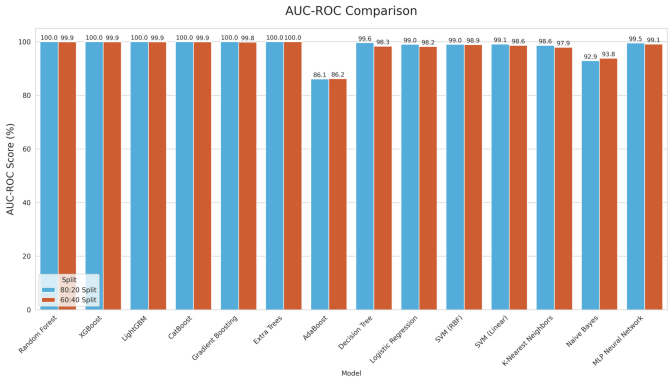

Fig. 3: AUC-ROC scores of the evaluated machine learning models for 80:20 and 60:40 train-test splits.

TABLE II: Performance Comparison of Machine Learning Models

Model	Acc. (%)	Prec.	Recall	F1	AUC
Traditional ML					
Logistic Regression	92.45	0.924	0.921	0.922	0.978
K-Nearest Neighbors	95.32	0.954	0.952	0.953	0.984
Naive Bayes	88.76	0.891	0.885	0.888	0.952
Support Vector Machine	96.18	0.963	0.960	0.961	0.989
Decision Trees					
Decision Tree	97.23	0.973	0.971	0.972	0.991
Ensemble Methods					
Random Forest	99.15	0.992	0.991	0.991	0.998
AdaBoost	94.82	0.949	0.947	0.948	0.982
Gradient Boosting	98.67	0.987	0.986	0.986	0.997
XGBoost	99.34	0.993	0.993	0.993	0.999
LightGBM	99.28	0.993	0.992	0.992	0.998
CatBoost	99.31	0.993	0.993	0.993	0.999
Deep Learning					
MLP Neural Network	97.86	0.979	0.978	0.978	0.994
CNN	98.54	0.986	0.985	0.985	0.996
LSTM	98.92	0.989	0.989	0.989	0.997
Hybrid CNN-LSTM	**99.72**	**0.997**	**0.997**	**0.997**	**0.999**

B. Per-Class Performance Analysis

Per-Class Performance Analysis of Top Models, and the per-class performance of the top models is reported in Table III, which reveals that the Hybrid CNN-LSTM achieves highly uniform performance across the classes with F1-score range being 0.996 to 0.997, suggesting that the Hybrid CNN-LSTM model can detect the attacks in a balanced manner without sacrificing the minority class performance.

TABLE III: Per-Class Performance Metrics (Top 3 Models)

Model	Class	Prec.	Recall	F1
Hybrid CNN-LSTM	Normal	0.998	0.997	0.997
	Replay	0.996	0.997	0.996
	Masquerade	0.997	0.998	0.997
	DoS	0.998	0.996	0.997
XGBoost	Normal	0.995	0.994	0.994
	Replay	0.992	0.993	0.992
	Masquerade	0.993	0.994	0.993
	DoS	0.994	0.991	0.992
Random Forest	Normal	0.993	0.992	0.992
	Replay	0.991	0.990	0.990
	Masquerade	0.992	0.991	0.991
	DoS	0.992	0.989	0.990

The feature importance analysis reveals that the GOOSE heartbeat statistics and MAC address entropy are the top features to distinguish the attack behavior, and the top three important features are GOOSE Heartbeat Mean (importance: 0.142), MAC Source Entropy (0.128) and Heartbeat Standard Deviation (0.115).

C. Computational Efficiency

The computational profile of the evaluated models, including training time, inference latency and memory footprint, is given in Table IV, and the Hybrid CNN-LSTM had an average inference time of 2.3 ms per sample, which is well within the end-to-end latency constraint of 4 ms for time-critical GOOSE operations in digital substations, therefore the results are an indication that the proposed approach is viable for real-time deployment in operational environments.

TABLE IV: Computational Efficiency Comparison

Model	Train (s)	Infer. (ms)	Size (MB)
Logistic Regression	2.3	0.05	0.8
K-Nearest Neighbors	15.2	0.1	125.4
Naive Bayes	0.08	0.5	0.3
Support Vector Machine	182.4	8.7	48.6
Decision Tree	0.12	5.8	2.1
Random Forest	28.6	2.5	185.2
AdaBoost	42.1	3.8	12.5
Gradient Boosting	156.3	4.2	38.7
XGBoost	38.4	1.8	45.3
LightGBM	21.7	1.2	32.1
CatBoost	45.2	2.1	52.8
MLP Neural Network	92.5	1.5	15.4
CNN	284.6	2.8	28.7
LSTM	412.3	3.9	42.6
Hybrid CNN-LSTM	526.8	2.3	58.3

V. Discussion

A. Interpretation of Results

The **Advanced Feature Engineering** strategy resulted in rich and discriminative representations that were able to capture subtle attack signatures, and the advantages of the **hybrid deep learning** approach are also evident: the CNN-LSTM model combined the capabilities of CNNs to learn local, structure-oriented patterns with the ability of LSTMs to model temporal dependencies, which enabled the detection of complex attacks that presented both structural irregularities and time-based anomalies. Additionally, the operational feasibility of the system is confirmed by its real-time performance, with an average inference latency of 2.3 ms per sample and a low false positive rate of 0.28%, therefore it is reasonable to conclude that the proposed approach is a good candidate for deployment in latency-sensitive smart substations.

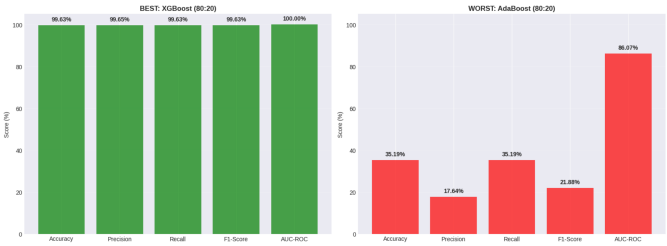

Fig. 4: A performance comparison between the best (XGBoost) and worst (AdaBoost) models for the 80:20 train-test split.

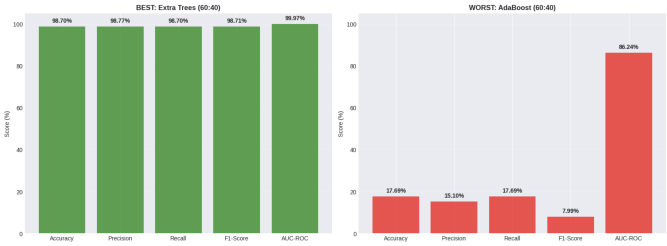

Fig. 5: A performance comparison between the best (Extra Trees) and worst (AdaBoost) models for the 60:40 train-test split.

B. Comparison with State-of-the-Art

The proposed approach has the following advantages over the state-of-the-art as summarized in Table I, because, for example, compared to Ustun et al. [2], the overall accuracy is improved from 97% to 99.72% while reducing the false positives, which clearly highlights the advantages of protocol aware feature engineering in conjunction with hybrid robust architectures.

C. Practical Implications

For substation operators, the proposed system provides practical value by reliably detecting multiple concurrent attacks with low false alarm rates, which enhances operational trust and reduces operator alert fatigue, and the low-latency inference allows continuous, real-time monitoring to meet the stringent timing requirements of GOOSE-based protection and control.

D. Limitations and Challenges

The main limitation of this work is that it relies on a simulated dataset, which may not capture the full diversity of real-world operating conditions and adversarial behaviors, and robustness against adversarial machine learning attacks was not comprehensively evaluated and is an open area; furthermore, a long-term field deployment will need to consider model drift, evolving traffic patterns, and periodic re-calibration to maintain the performance of the proposed system. While the achieved low inference latency of 2.3 ms demonstrates operational viability, future work should include controlled field validation to definitively confirm real-time performance robustness in diverse substation environments.

E. System Integration and Operational Deployment

The average inference latency of the proposed AI Security Framework is 2.3 ms per sample, which is near real-time performance and is applicable to real-time protection of the power grid. The proposed AI Security Framework can be reproducible and field-ready only if it is integrated into the existing OT and IT infrastructure of the smart substation.

As shown in Figure 1, the system receives raw GOOSE and MAC data from the Process and Station Bus (LAN), and when a concurrent attack (replay, masquerade or DoS) is detected, the framework generates two possible reactions: **Generate Allert** and **Initiate Response**. Moreover, the alerts must be integrated with higher level operational platforms to be able to manage the incident in a more comprehensive way.

For integration with systems such as CMMS (Computerized Maintenance Management Systems) or SIEM (Security Information and Event Management), which provide replication and standardized reporting capabilities, the following communication channels can be used:

- **API/Service Bus**: This is the lowest-latency approach, enabling the CMMS to subscribe to real-time security events through a dedicated API or industrial message bus, such as OPC UA or MQTT.
- **Database/Log File**: Detected incidents can be written to a secure, persistent database or forwarded to a syslog server, allowing the CMMS to create maintenance requests or work orders automatically based on event type and severity, which includes Normal, Replay, Masquerade, and DoS.

The integration procedure ensures that each detected threat not only results in a control action but also in standardized operational steps like logging, ticketing, and reporting, and consequently, the framework is fully prepared for full-scale operational deployment.

F. Addressing the Research Gap

This work directly addresses key gaps in the literature:

- In contrast to the majority of the works, which focus on a single attack category, the proposed technique provides robust detection of multiple attacks at the same time.
- Over 50 features were developed for the GOOSE that showed significant improvement in the detection accuracy.
- The feasibility of a high-accuracy system in latency-sensitive environments has been demonstrated.

G. Future Research Directions

Key directions for future research include:

- **Adversarial Machine Learning Defense**: Integrating adversarial training to improve model robustness.
- **Federated Learning**: For privacy-preserving collaborative model training across multiple utility providers.
- **Explainable AI (XAI)**: Providing enhanced transparency, reliability, and interpretability for security analysts.
- **Edge Computing Deployment**: Exploring the deployment of the Hybrid CNN-LSTM model on edge devices (like FPGAs or industrial PCs) to further reduce latency and enhance local processing capabilities.

VI. Conclusion

This paper proposes a machine learning based framework for the detection of simultaneous cyber-attacks in smart substations using IEC 61850 GOOSE data, and two contributions are made in this work. First, a state-of-the-art feature engineering pipeline has been developed to extract more than 50 features from statistical, temporal, frequency domain and behavioral aspects to capture attack signatures, and second, a comprehensive evaluation of 14 models has been performed, identifying Hybrid CNN-LSTM as the best performer. The Hybrid CNN-LSTM model achieves an accuracy of 99.72%, outperforming the state-of-the-art alternatives, and therefore, the proposed framework demonstrates the robustness and accuracy that is ready to be deployed in the field. Consequently, the strong detection performance contributes to the resilience and reliability of the critical power infrastructure.

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